

MUM 2026 · MATSim User Meeting

A multi-source calibration framework for activity-based transport simulations

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28/09/2026

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Simulating Mobility for Policies Evaluation

Multi-agent simulations are increasingly used to support public **decision-making**.

- They enable the analysis of prospective “**what-if**” scenarios.

Motivation:

Request from the City of Paris: develop an activity-based multi-agent simulation model to test different scenarios before implementation.

- Limited Traffic Zones,
- Reserved lanes,
- Low-Emission Zones,
- Parking spaces,
- ...

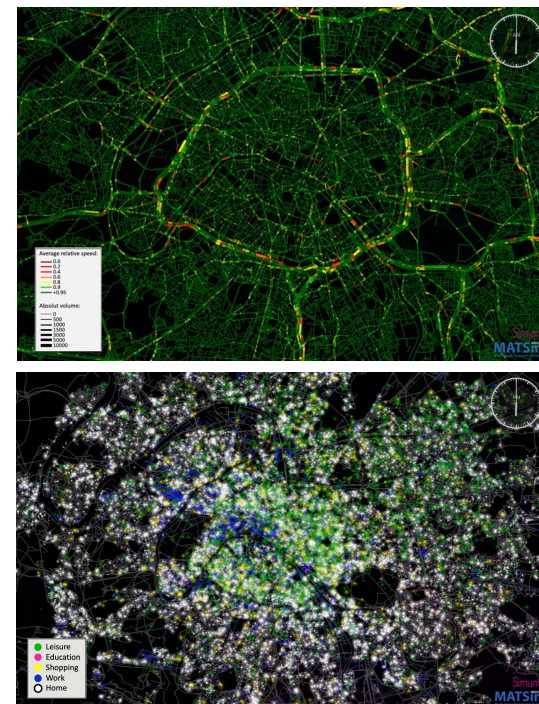


Figure1: Simulation multi-agents de l'Île-de-France (MATSim)

State of the Art in Calibration

1 Supply

Network capacity & free-flow speed

- *Free flow speeds using Google API — Melbourne [16]*
- *Time dependent capacity and speed (INRIX) [9]*
-

2 Demand

Choice parameters: mode, destination, time, route

- *Cadyts [7, 8] · SimMobility [3]*
- *TRANSIMS [18] · DCM–MATSim, Zurich [11]*
- ...

3 Simulation parameters

Replanning, time step, iterations

- *Sensibility analysis — Pekin [19]*
- ...

Multi-step approaches

- *Barcelone — Toll, validation on 11 axes [4]*
- *Open Berlin — Activity locations using Cadyts[20] ; extension SUMO + fret [14]*
-

Limitations of Calibration Methods

- Partial calibration (only one class of parameters)
- Methods not suited to all parameters (e.g. Cadyts)
- Only aggregated calibration targets (e.g. mode shares, daily traffic volumes)
- Incomplete demand representation (freight, services, visitors)



Unrealistic traffic
conditions

+

Limited transferability to future scenarios

Objectives:

1. Empirically demonstrate that aggregate calibration alone is not sufficient to reproduce observed traffic conditions
2. Develop a multi-source calibration framework to improve traffic realism

Case Study: Paris

- **Study area:** Île-de-France /Paris
- **Simulation platform:** MATSim
- **Simulated population:** 123 178 habitants
- **Sample:** 1 % of the population of Île-de-France
- **Hourly traffic count data:** 1 892 loop detectors
- **Travel time data:** TomTom

- Detector–link matching based on: overlap, distance, road name, direction, and plausibility
- Resident synthetic population generated using the eqasim framework [10], EGT 2020

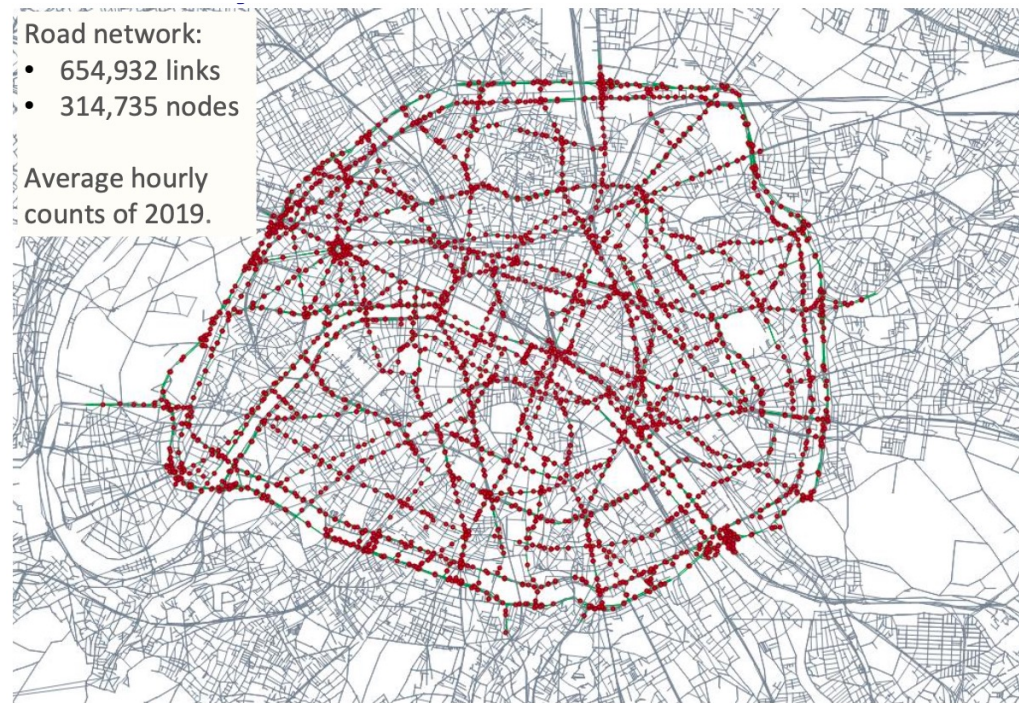


Figure 2: Spatial distribution of traffic count detectors on the Paris road network.

Calibration Framework: Four Steps

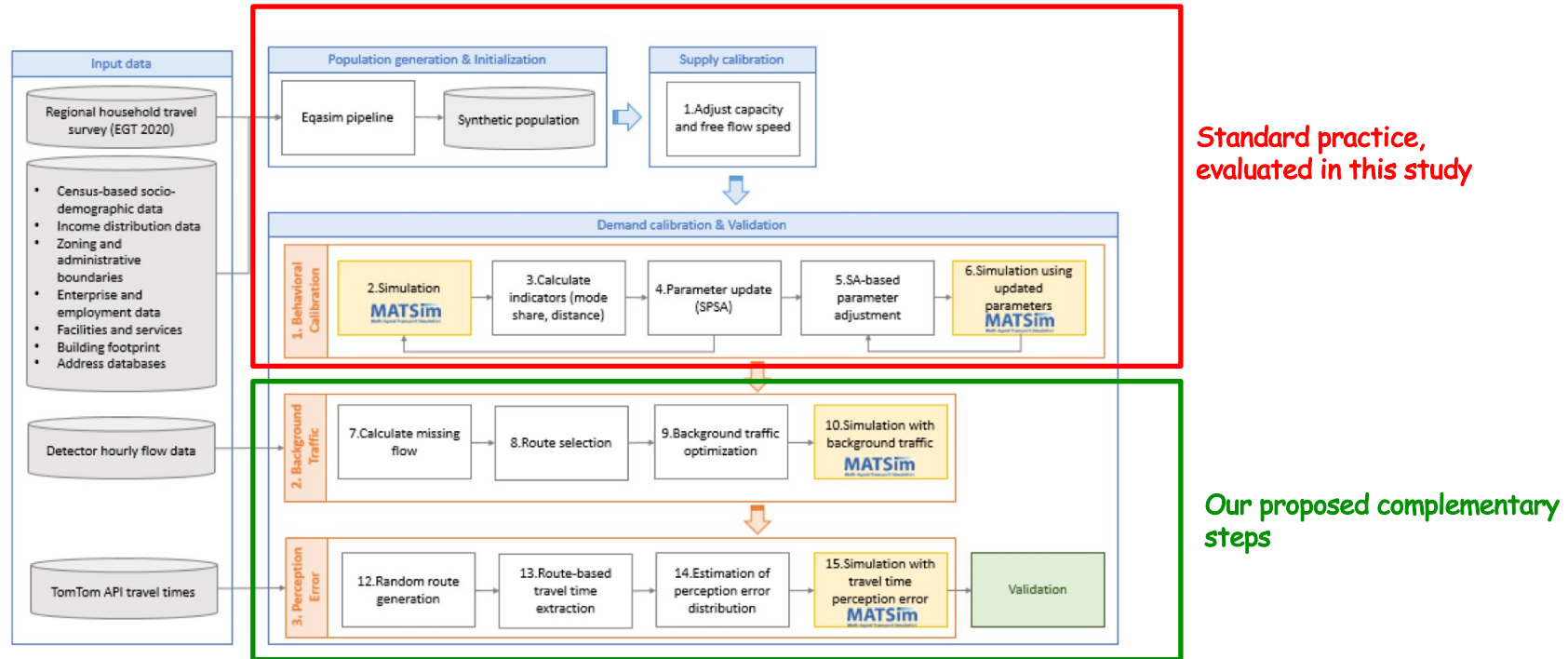
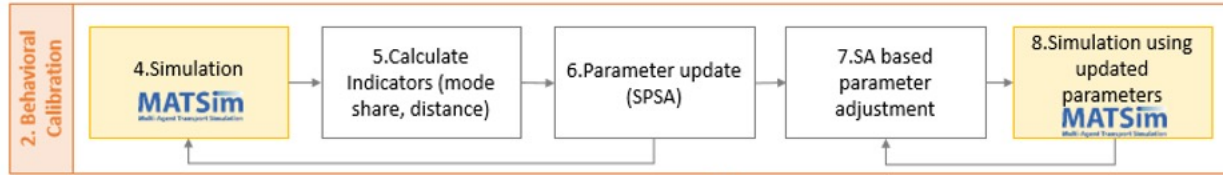


Figure 3: General overview of the proposed calibration framework.

Behavioral calibration



Objective: Modal share per distance category (EGT 2020)

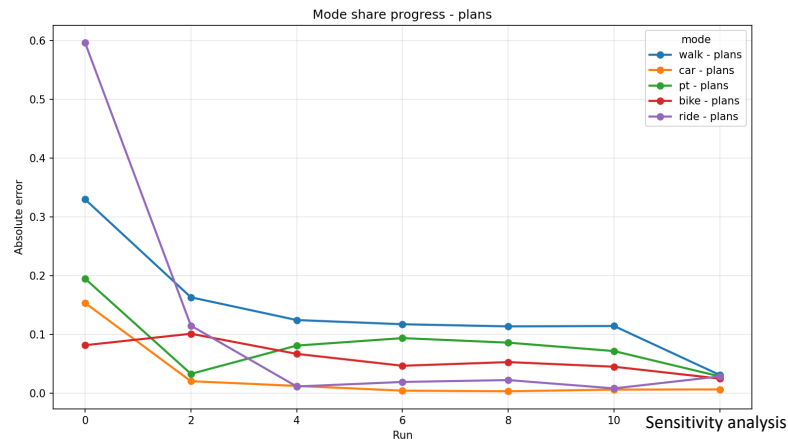


Figure 6: Mode share error per iteration

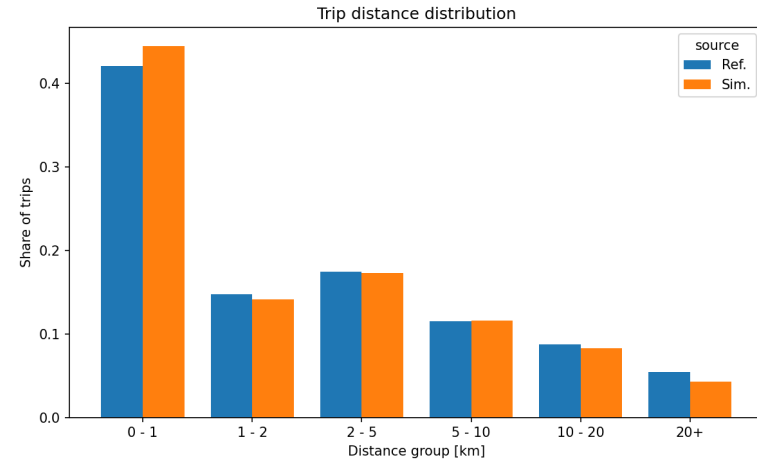


Figure 7: Trip distance distribution

Objective 1:

Is matching aggregate measures sufficient to reproduce realistic traffic conditions?

Good agreement in modal share, BUT traffic is underestimated

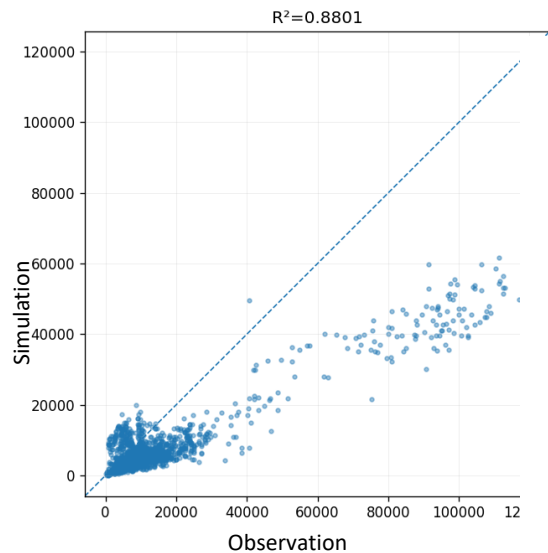


Figure 8: Comparison of simulated and observed daily traffic

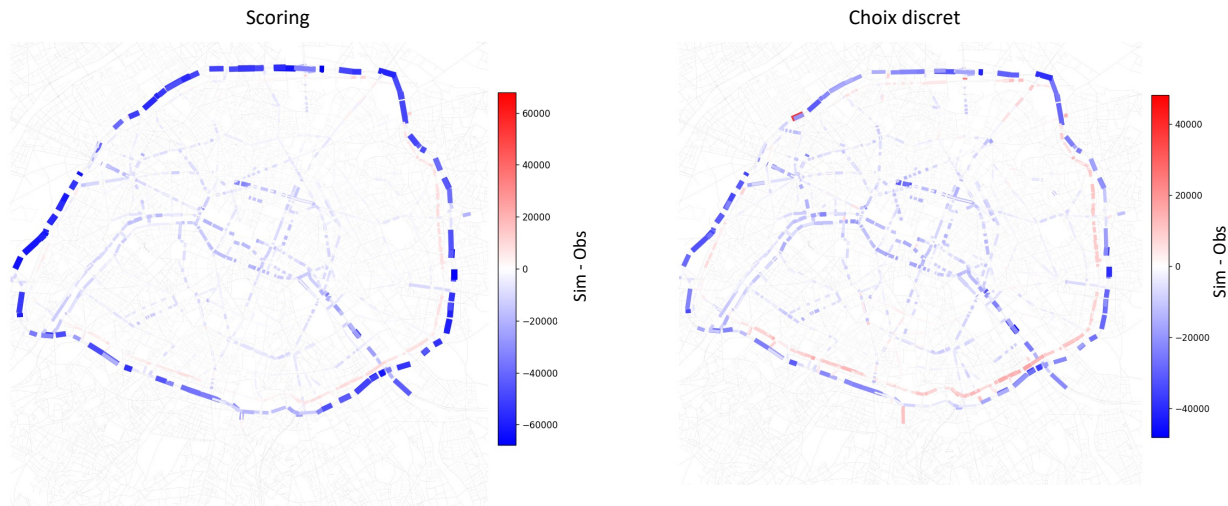


Figure 9: Difference between simulation and observations — daily traffic

➤ Behavioral consistency does not guarantee realistic traffic

Objective 2:

How can **traffic realism** be improved?

The gap between simulated and observed traffic can be explained by several factors, including:

- **Unrepresented demand: freight, service vehicles, visitors, and non-residents**
- **Agents' perception errors**
- ...

Reconstructing Missing Demand



- The EGT mainly represents resident travel, leaving freight, service trips, and visitor travel underrepresented.



Background-flow estimation:

$$\min \| A_h x_h - \Delta_h \|^2 + \lambda \| x_h - x_{0,h} \|^2$$

Fit the observed
missing flows

Stay close to a
prior OD
matrix (low λ)

$$\text{sous } x_h \geq 0 \cdot A_h x_h \leq b_h$$

Δ_h : residuals

A_h : incidence matrix

λ : regularization parameter

- Positive residuals (observed – simulated) are converted into agents with fixed routes
- Capacity constraints are applied to avoid network blocking

Reconstructing Missing Demand



Background traffic: a more dispersed daily profile

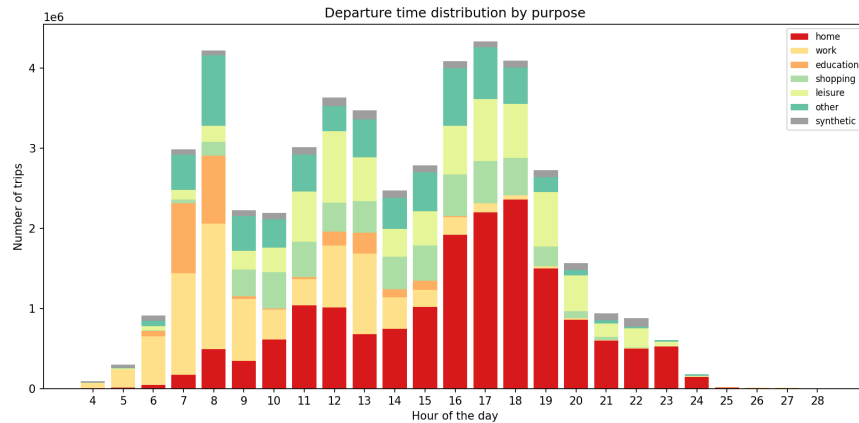


Figure 12: Distribution of departure times by trip purpose (including background traffic)

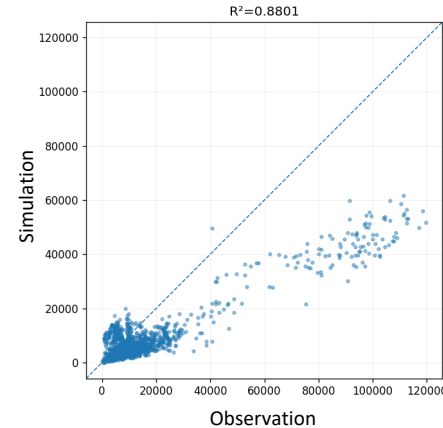


Figure 13: Without background traffic — $R^2=0.88$

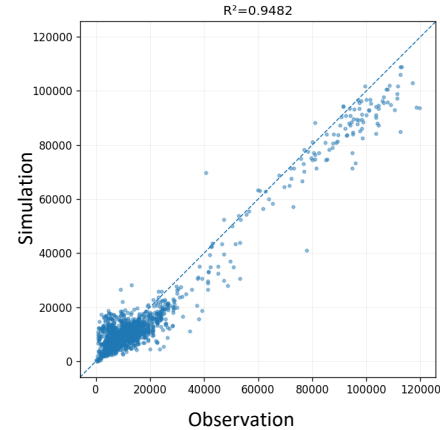
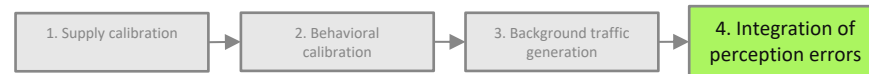


Figure 14: With background traffic — $R^2=0.95$

Table 1. Detector-count fit before and after background-traffic integration across aggregation periods.

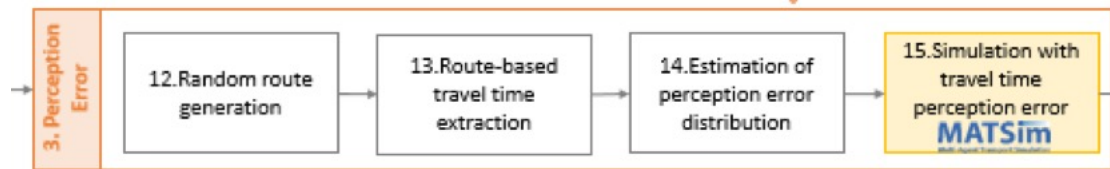
Period	Before background traffic			After background traffic		
	R^2	NRMSE	MAE	R^2	NRMSE	MAE
Day	0.897	0.738	7,001	0.946	0.334	3,769
AM peak	0.856	0.599	1,451	0.924	0.379	997
PM peak	0.844	0.552	1,534	0.917	0.371	1,072
Hourly	0.669	0.845	483	0.887	0.454	294

Accounting for Perception Errors



- In the simulation, agents select the plan with the highest score.
- The standard scoring function assumes perfect perception of travel time.

→ *Excessive concentration on “optimal” routes.*



Perception error, ε , is estimated from the difference between **MATSim** and **TomTom** travel times.

$$\tilde{T}_i = T_i + \varepsilon_i$$

Travel context

- OD: Paris–Paris, External–Paris, External–External
- Period: morning peak / evening peak / off-peak
- Distance within Paris: 0–2, 2–5, 5–10, 10–15 km

$$S_{plan} = S_{activity} + S_{strip}$$

$$\beta_1 * (travel\ time + \varepsilon) + \dots$$

Accounting for Perception Errors

Agent-based Paris subroutes sampled by distance class



Around 800 routes are used every 15 minutes.

Travel-time accuracy before and after perception-error calibration ($\eta = 3$), evaluated on 832 routes.

Level	Granularity	MAE (min)	RMSE (min)	MAPE (%)
Network	Daily	3.24 → 2.42	3.24 → 2.42	28.3 → 21.2
Network	15-min	3.23 → 3.03	4.02 → 3.75	32.1 → 30.3
Route	15-min	5.92 → 3.61	10.07 → 5.99	66.4 → 63.5

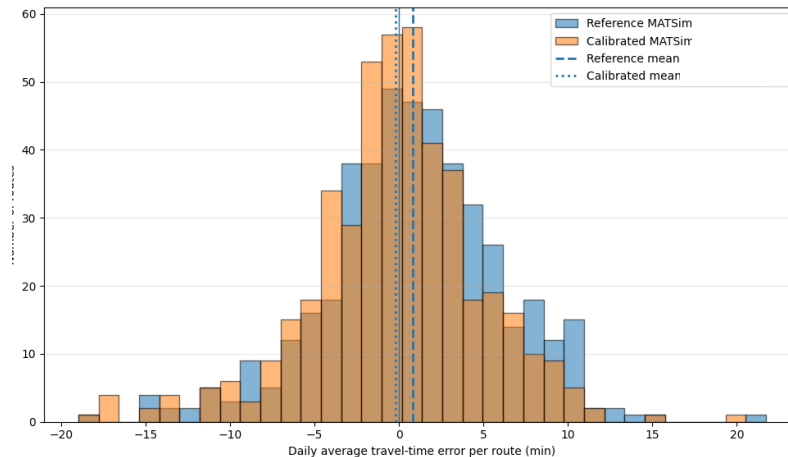
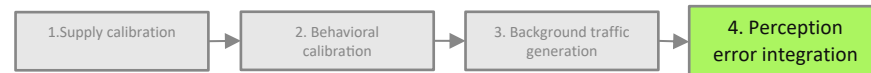


Figure 16: Route travel-time error compared with TomTom

Conclusion

- Aggregate indicators are necessary — but they are not sufficient to ensure realistic traffic conditions.
- Multi-source calibration framework (survey + traffic counts + travel times):
 - Supply calibration
 - Demand calibration
 1. Calibration of choice parameters
 2. Background traffic
 3. Perception error

Limitations:

- Sampling
- Validation data

Next steps:

- Testing the transferability of the calibrated model to future scenarios

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Thank you for your attention



9–11 June 2027, Université Gustave Eiffel, Champs-sur-Marne, France